```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(
    "/data.txt",
    sep=r"\s+",
    engine="python"
)

df.head()
```

	time	V(n002)	I(V1)
0	0.000000	145.6878	-3.890075e-05
1	0.000002	145.5795	2.529091e-07
2	0.000006	145.3629	2.529450e-07
3	0.000014	144.9297	2.529746e-07
4	0.000030	144.0633	2.530751e-07

Next steps: New interactive sheet

```
time = df["time"].values
signal = df["V(n002)"].values
```

```
dt = np.diff(time)

print(dt.min())
print(dt.max())
```

```
8.313561465078667e-07
2.000000000010327e-05
```

```
Fs = 50000          # choose sampling frequency (Hz)
dt = 1/Fs

t_uniform = np.arange(time[0], time[-1], dt)

signal_uniform = np.interp(
    t_uniform,
    time,
    signal
) #interpolated signal
```

```
time = t_uniform
signal = signal_uniform
```

```
signal = signal - np.mean(signal) # dc offset removal
```

```
N = len(signal)

fft = np.fft.rfft(signal)

freq = np.fft.rfftfreq(N, d=1/Fs)

magnitude = np.abs(fft) * 2 / N
```

```
plt.figure(figsize=(12,5))

plt.plot(freq, magnitude)

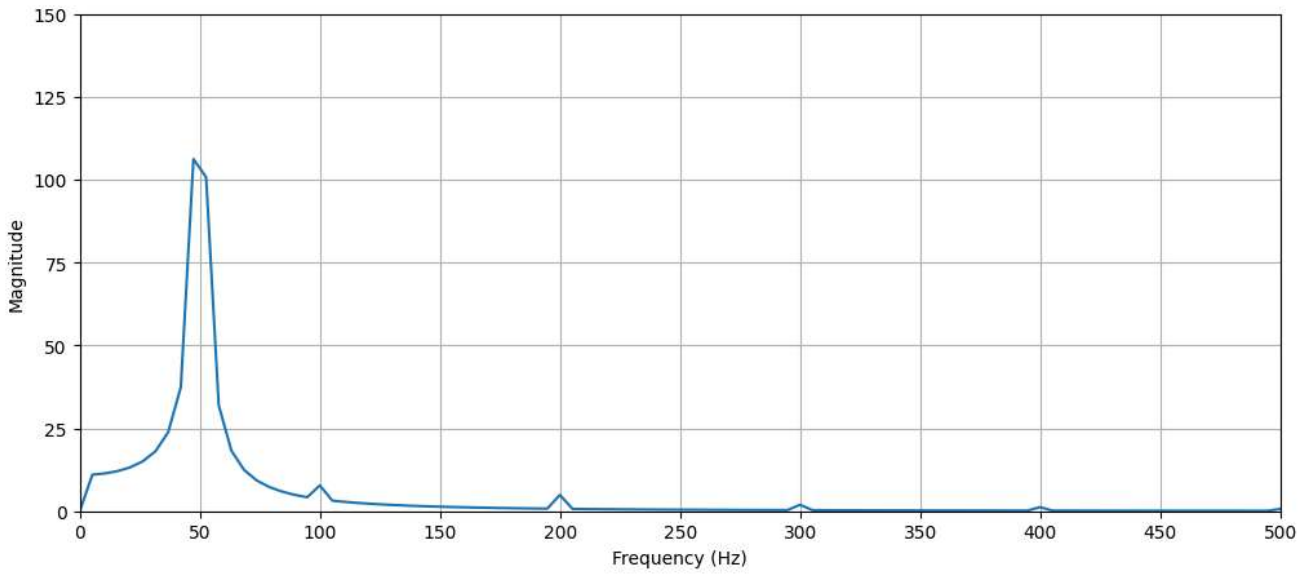
plt.xlim(0,500)
plt.xticks(np.arange(0,501,50))
plt.xlabel("Frequency (Hz)")

plt.ylabel("Magnitude")
plt.ylim(0,150)
```

```
plt.yticks(np.arange(0,151,25))

plt.grid(True)

plt.show()
```




```
fundamental = 50

orders = np.arange(1,21)

harmonics = []

for order in orders:

    target = order*fundamental

    idx = np.argmin(np.abs(freq-target))

    harmonics.append(magnitude[idx])

harmonics = np.array(harmonics)
```

```
for order, amp in zip(orders, harmonics):
    print(
        f"H{order:2d}  "
        f"{order*fundamental:6.1f} Hz  "
        f"{amp:.4f}"
    )
```

```
H 1    50.0 Hz  106.2463
H 2   100.0 Hz  7.7693
H 3   150.0 Hz  1.4167
H 4   200.0 Hz  4.9107
H 5   250.0 Hz  0.4639
H 6   300.0 Hz  1.9372
H 7   350.0 Hz  0.2304
H 8   400.0 Hz  1.2027
H 9   450.0 Hz  0.1378
H10   500.0 Hz  0.6915
H11   550.0 Hz  0.0917
H12   600.0 Hz  0.3050
H13   650.0 Hz  0.0654
H14   700.0 Hz  0.2975
H15   750.0 Hz  0.0490
H16   800.0 Hz  0.2952
H17   850.0 Hz  0.0381
H18   900.0 Hz  0.1731
H19   950.0 Hz  0.0304
H20  1000.0 Hz  0.1464
```

```
plt.figure(figsize=(10,4))
```

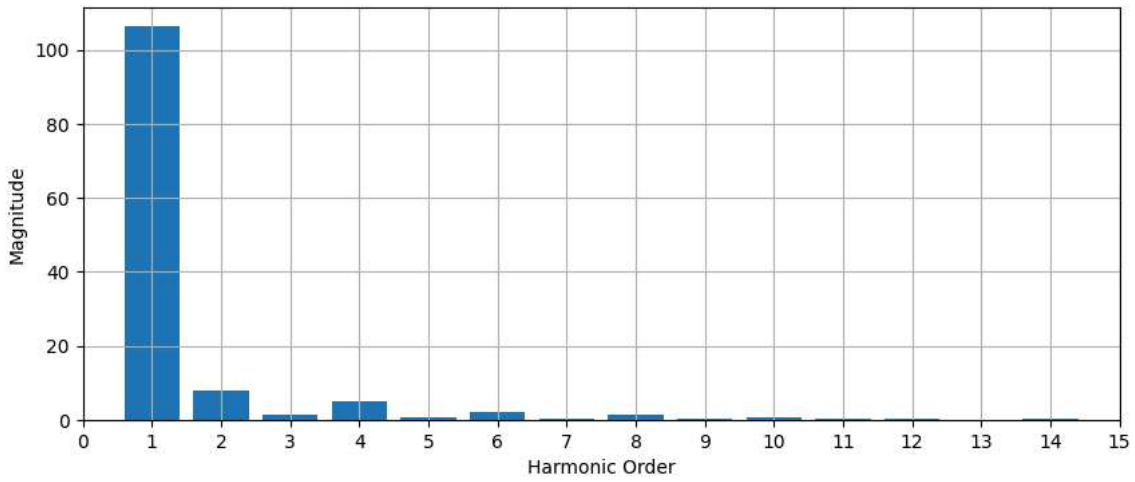
```
plt.bar(orders, harmonics)

plt.xlabel("Harmonic Order")
plt.xlim(0,15)
plt.xticks(np.arange(0,16,1))

plt.ylabel("Magnitude")

plt.grid(True)

plt.show()
```




```
V1 = harmonics[0]

THD = np.sqrt(np.sum(harmonics[1:]**2))/V1*100

print(f"THD = {THD:.2f}%")
```

```
THD = 9.07%
```